DALHOUSIE UNIVERSITY

Psychology and Neuroscience 46th Annual Graham Goddard In-House Conference

May 27, 2022
Life Sciences Centre
Room 4260

IN HOUSE SCHEDULE – FRIDAY, MAY 27TH

TALKS
(All talks will be held in
LSC Psychology, Room 4260 on the 4th floor)

9:15-10:15 Session 1 Chair: Hélène Deacon

T1	Aaron Newman	Welcome to the Conference
T2	Samantha Good	Examining Autism Spectrum Disorder using the Attention Network Test: A Meta-Analysis
T3	Saisha Rankaduwa	Re-analysis of body representation in the somatosensory cortex: An investigation of the “femunculus”
T4	Helene Deacon	Reading through the elementary school years, with a break for COVID

10:15-10:45 Coffee Break

10:45-11:30 Session 2 Chair: Sean Mackinnon

T5	Colin McCormick	Lifestyle changes that can impact your attention
T6	Luke Mungall	Personality, Political Ideology, and Partisanship in Canada
T7	Sean Mackinnon	A History of Grading Practices at Dalhousie University

11:30-12:00 STUDENT JEOPARDY!

12:00-1:00 Lunch - 2nd Floor Lounge *(or outside weather permitting. Note that people will have to disperse to eat and cannot eat in the poster session itself.)*

12:30-2:00 Poster Conference - 4th Floor Lilyan E. White Study Centre *(Masks must be worn at all times.)*

2:00-3:30 Session 3 Chair: Tara Perrot

T8	Oliver Schnare	Measuring Memory Strength in 5xfad Mice Through Olfactory Discrimination Tasks and *Atrx* Gene Expression in Hippocampus
T9	Dylan Miller	Hemolymph Nutrient and Metabolite Changes in Caterpillars Experiencing Parasite-Induced Feeding Suppression
T10	Nicole Michaud	Measuring Size Perception in Unrestrained Wild-type Mice Using a Novel Two-Alternative Forced Choice Behavioural Task
T11	Christopher Makau	Inhibition of Nav1.7 and Nav1.8 Decreases Pain-Related Behaviour in Speke’s Hinge-Back Tortoise (*Kinixys Spekii*)

T12	Elizabeth O'Leary	Acute Probiotic Treatment During the Perinatal Period Affects Maternal Care and Gut Microbiota Profiles of Long-Evans Rats
T13	Brodie Babcock-Parks	Natural Photoperiodic Regulation of Hippocampal Plasticity in Wild Black-Capped Chickadees (*Poecile Atricapillus*)

3:30-4:00 Coffee Break

4:00-5:30 Session 4 Chair: Aaron Newman

T14	Tara Perrot	Improvements to Rodent Housing Have Significant Effects on Development of the Stress Response: Implications for Modeling Neuropsychiatric Disorders
T15	Ray Klein	Is There One "Beam" of Attention for Searching in Space and Time?
T16	Jessica Garden	Pavlovian Conditioned Odour Memory in the *Neurexin1+/-* Mouse Model of Autism Spectrum Disorder
T17	Richard Drake	Reconciling Expectation and Interpretation in the Cognitive Training Meta-Analyses
T18	Laura Elliott	Lexical Decisions in Ortho-Semantic Learning: A behavioural Study of Grade 3 Children in English and French Immersion
T19	Keira O'Neil	The Visual Word Form Area as a Route of Auditory Linguistic Information Into the Visual Cortex of Early Blind Individuals: A Dynamic Causal Modelling FMRI Study
	Aaron Newman	Concluding Remarks

POSTERS

(All posters will be presented from 12:30 pm to 2 pm in the Lilyan E. White Study Centre on the 4th floor)

P1	Brenna Bagnell	Perinatal Grief as a Mediator Between Religiosity and Sexual Satisfaction Among Couples Who Have Experienced Pregnancy Loss
P2	Mitch Baker	Investigating Retinal Inactivation as a Safe Therapy for Amblyopia
P3	Celia Briand	The Power of the Pencil: Examining the impact of drawing on learning and memory
P4	Dante Cantini	Implications of Lactobacillus Probiotic Administration and Maternal Western Diet on Active Maternal Care and Offspring Development in Rats
P5	Eva Cohen	A Dyadic Comparison of Sexual Desire and Sexual Satisfaction Between Those Who Have Experienced Pregnancy Loss and Those Who Have Not
P6	Laura Dauphinee	Investigation of the Programming Effects of Probiotic Treatment on Metabolic and Anxiety-Related Outcomes in Male and Female Long-Evans Rats
P7	Bayla Dolman	The role of strategic correction in prism adaptation effects
P8	George Faza	Personality, Stress, and Academic Cheating During the COVID-19 Pandemic
P9	Brett Feltmate	Examining “Immunity” From Attention Capture: Does Cueing Fixation Delay Disengaging From It?
P10	Jonathan Henneberry	The Tale of Two Antibodies: A NESC 3370 and NESC 3000 Project
P11	Chris Holland	Automation Reliability and Its Influence on Trust Dynamics
P14	Tess Konwisarz and Colin Conrad	Investigating the Impact of Pre-recorded Lecture Length on Mind Wandering and Learning Outcomes
P15	Ryan Lewis and Tracy Taylor-Helmick	Interaction of memory and attention in the context of directed forgetting
P16	Madison MacLachlan	The effect of prosocial behaviours on preschooler’s accent processing and word learning
P17	Cory Munroe	The Factor Structure of a New Executive Function Questionnaire in Young Adults: The Learning, Executive, and Attention Functioning (LEAF) Scale
P18	Nicholas Murray	Is Gaze-Cuing More Like Endogenous or Exogenous Orienting?
P19	Amber Myatt	How COVID-19 Restrictions Impacted Uptake of the Better Nights, Better Days Program

P20	Anika Nastasiuk	Does Physical Activity Predict Executive Function in University Students?
P21	Hailey Silver	Relationship Between Satisfaction and Willingness to Pay for the Better Nights, Better Days for Children with Neurodevelopmental Disorders Program
P22	Aaron Shephard	The impact of anxiety sensitivity on self-esteem and negative affect following a statistics test
P23	Jeff Zbarsky	The Role of Muscle in the Immune Response: A RT-Qpcr Study of Toll1 and Insulin Receptors in the Muscle and Fat Body of *Manduca Sexta*
P24	Radostina Zhekova	The Effects of Cannabidiol Expectancy on Acute Measures of Stress and Anxiety

TALKS
(All talks will be held in
LSC Psychology, Room 4260 on the 4th floor)

T1) TALK TITLE: Welcome to the Conference
Aaron Newman

T2) TALK TITLE: Examining Autism Spectrum Disorder Using the Attention Network Test: A Meta-Analysis

AUTHORS: Samantha Good, Swasti Arora, Jeanne Townsend, Lisa Mash & Raymond Klein

ABSTRACT: Attentional deficits are common in autism spectrum disorder (ASD) however the nature of these deficits is yet to be fully understood. The present study aimed to assess the three attention networks (alerting, orienting, and executive control) as identified by Posner and Peterson. We conducted a Bayesian hierarchical meta-analysis of studies that implemented the Attention Network Test or its variants to explore whether individuals with ASD exhibited differences in any of the attentional networks when compared to typically developing controls. Eight studies were included in our analysis, comprising 322 children and young adults with the average age ranging from 4.76 - 30 years (*Mdn* = 13.24). There were no credible differences between participants with ASD versus typically developing controls in the alerting network. Orienting scores were not credibly different, however there was a trend suggesting less efficient attentional response to spatial information in ASD. Executive control scores were not credibly different, but there was a trend showing higher average scores in people with ASD. Although the results from this meta-analysis add to the growing literature on the nature of attentional deficits in people with ASD, our findings are limited by the relatively small number of studies.

T3) TALK TITLE: Re-Analysis of Body Representation in the Somatosensory Cortex: An Investigation of the "Femunculus"

AUTHORS: Saisha Rankaduwa & Jennifer Stamp

ABSTRACT: The original 2-D somatosensory homunculus based on Penfield's work lacks representation of uniquely female anatomy. For this reason, we created a quantitative 2-D map of female-specific anatomy in the primary somatosensory cortex (S1) based upon a systematic review of published neuroimaging literature, that comprised 11 studies. These studies reported localization of representations of the clitoris, cervix, vagina, dorsal pudendal nerve, perineal nerve, nipple, and breast in the medial wall of S1 in the paracentral lobule. Representations of the dorsal clitoral nerve, posterior vulva, clitoris, breast, and nipple were also found in the secondary somatosensory cortex. Because there were inconsistencies between the locations of anatomy between studies, we proposed a research plan that comprehensively maps female specific anatomy in the somatosensory cortices. The creation of a female somatosensory "femunculus" is necessary considering the implications it has for assessing neuroplasticity after changes to female anatomy in mastectomies and gender affirmation surgery.

T4) TALK TITLE: **Reading Through the Elementary School Years, With a Break for COVID**

AUTHORS: Hélène Deacon, Stef Hartlin, Alex Ryken, Mariam Elgendi, Elizabeth MacKay, Klaudia Krenca & George Frempong

ABSTRACT: In 2019, we optimistically launched a longitudinal study with over 300 children in Grade 1 at 18 schools around the province. In partnership with the Dep't of Education, our goal was to follow them through to Grade 6. But, of course, COVID hit when the children were in Grade 2, midway through testing. I report here on our findings from our first couple of years and how we are working to recover from this change in plans. The oral language skills that we report here on an IURN and SSHRC-funded research study designed to investigate the language skills that children use in their reading. We recruited over 300 elementary school aged children, initially tested in Grade 1 (in winter of 2019). We report here on analyses exploring the relations between early language skills and Grade 2 reading levels. We also explore the achievement and opportunity gap in Nova Scotia by contrasting results for African Nova Scotians with those for other Nova Scotian children. Our goal is to provide another lens into the skills that support all children in learning to read.

T5) TALK TITLE: **Lifestyle Changes That Can Impact Your Attention**

AUTHORS: Colin R McCormick

ABSTRACT: This talk will summarize a literature review on how different lifestyle factors impact the networks of attention. With the help of the 'ANT Database' developed in the Klein Lab (found at AttentionNetwork.ca), 70+ studies were included that fit into five main categories: meditation, exercise, drug use, sleep, and environmental or social factors. All three networks showed that they were sensitive to different changes in lifestyle. This review also reveals gaps in the literature for understanding how attention is being impacted by our day-to-day choices, as well as the importance of consistent reporting of results and the availability of experimental data.

T6) TALK TITLE: **Personality, Political Ideology, and Partisanship in Canada**

AUTHORS: Julie Blais, Luke R. Mungall & Scott Pruysers

ABSTRACT: Personality traits reliably predict political ideology and partisanship around the world, but studies investigating these relationships among Canadians used poor measures of personality, or did not examine the contribution of antagonistic personality traits (i.e., psychopathy, narcissism, Machiavellianism). We surveyed Canadians (N = 2551) on their ideological self-placement and political partisanship using robust measures of HEXACO traits and antagonistic traits. The findings for political ideology were mostly consistent with previous studies. Canadians who placed themselves on the ideological right were lower in openness to experience, honesty-humility, and impulsivity, but higher in extraversion, narcissism, antisocial tendencies, and blunted affect. Nuanced relationships emerged between personality and support for specific political parties (i.e., partisanship). The personality traits of Liberal, Conservative, and Green partisans were not always consistent with the personality traits of their party's ideological wing. Overall, we demonstrated a crucial distinction between ideology and partisanship, and provided further evidence that personality matters for political choice in Canada.

T7) TALK TITLE: **A History of Grading Practices at Dalhousie University**

AUTHOR: Sean P. Mackinnon

ABSTRACT: This session will provide a historical overview of grading practices at Dalhousie. In the early 1900s, Dalhousie was using a variant of the British system, where 65% or higher was required for "distinction." By the late 1940s, additional subdivisions were created (First, Second, and Third Divisions), and the percentage grade cutoff for the highest grade moved to 80%. The 1970s were an experimental phase, where percentage grades were officially abolished and replaced with letter grades. After a period of virtually no standardization in grading across departments, the university moved towards standardization (i.e., the 4.3 GPA system and standardized percentage-to-letter conversion schemes) by present day. Hopefully, the historical context of how modern grading came to be will help us create new approaches that optimally meet both needs. Preprint: https://edarxiv.org/fmgv8/

T8) TALK TITLE: **Measuring Memory Strength in 5xfad Mice Through Olfactory Discrimination Tasks and *Atrx* Gene Expression in Hippocampus**

AUTHORS: Oliver Schnare & Dr. Richard E. Brown

ABSTRACT: We investigated behavioural and epigenetic signs of olfactory learning and memory in 5xFAD model mice and wildtype mice (9-16 months of age). Mice were classically conditioned on a single day using methods adapted from Schellinck et al. (2001) to associate the CS+ odour pot with a sugar reward, and the CS- odour pot with no reward. The behavioural signs of learning and memory were measured as their preference for the CS+ odour pot on 24-hour and 7-day memory tests. Epigenetic signs of learning and memory were measured as the normalized gene expression of *Atrx* in the hippocampus obtained from RT-qPCR analysis. Previously, mice with reduced *Atrx* expression display impairments to long-term potentiation in the hippocampus. 5xFAD mice spent significantly more time digging in the CS+ odour pot than wildtype mice during the acquisition stage, however their performances on memory tests did not differ. Wildtype mice displayed higher levels of *Atrx* expression than 5xFAD mice in the experimental groups, however *Atrx* expression served minimal predictive value to performance on memory tests overall. Our results suggest that 5xFAD mice do not have impairments to olfactory learning and memory, and that *Atrx* expression in the hippocampus may not be related to olfactory learning and memory capabilities.

T9) TALK TITLE: **Hemolymph Nutrient and Metabolite Changes in Caterpillars Experiencing Parasite-Induced Feeding Suppression**

AUTHORS: Miller, D.W., Zbarsky, J., Cohen, A.M., Penny, S. & Adamo, S.A.

ABSTRACT: Parasitic wasp larvae (*Cotesia congregata*) permanently suppress host caterpillar (*Manduca sexta*) feeding via an unknown mechanism(s) as the larvae emerge. This study investigated whether any changes in caterpillar hemolymph after wasps emerge may help explain this feeding behaviour change. Plasma samples were collected from five groups of caterpillars. Two control groups (feeding unparasitized caterpillars and food deprived caterpillars the parasitized non-feeding states) were compared to three groups of parasitized caterpillars: before wasp emergence, 1, and 3 days after emergence. Plasma metabolomics (via liquid chromatography with tandem mass spectrometry) identified nearly 200 metabolites in all five groups, including amino acids, nitrogenous waste, lipid

metabolism products, and immune signaling molecules (i.e., adenosine). The same groups were also assayed for sugar content-elevated hemolymph nutrients in the hemolymph suppress caterpillar feeding. These found that post-emergent caterpillars tended to have lower levels of sugars and amino acids compared to controls. The metabolomics also suggested an elevated immune activation signal for at least 3 days following parasite emergence. Immune activation can temporarily reduce or suppress feeding, and adenosine (immune signaling molecule) injection suppressed feeding for 2 hours in healthy caterpillars. These results provide insight into how wasps may be influencing their host's behaviour.

T10) TALK TITLE: Measuring Size Perception in Unrestrained Wild-type Mice Using a Novel Two-Alternative Forced Choice Behavioural Task

AUTHORS: Michaud, Nicole M. & Crowder, Nathan A.

ABSTRACT: Due to the availability of genetic tools used to explore neural circuits within and between brain areas, the mouse has become an invaluable animal model in vision research. However, to link neural circuits to visual perception, novel behavioural tasks must first be developed. To begin investigating the neural underpinnings of visual size perception, we modified the steering wheel paradigm developed by Burgess et al. (2017) to create a two-alternative forced choice size discrimination task in unrestrained wild-type mice. We assessed the ability of mice to correctly select the larger of two simultaneously presented grating patches, which allowed the generation of psychometric functions from perceptually driven behaviour. Mice displayed successful learning and generalization of the task when the reward stimulus size remained constant. However, when the size of the reward stimulus was varied and mice were required to select the relatively larger grating patch, task accuracy failed to reach criterion.

T11) TALK TITLE: Inhibition of Nav1.7 and Nav1.8 Decreases Pain-Related Behaviour in Speke's Hinge-Back Tortoise (*Kinixys Spekii*)

AUTHORS: Christopher M. Makau

ABSTRACT: Voltage-gated sodium ion channels in the peripheral nervous system have great potential as targets for eliminating pain with minimal side effects. Highly selective ion channel blockers, have been used to explore the involvement of a range of ion channels in nociception. Comparative studies are important to fully elucidate performance of such molecules and physiological mechanisms of particular ion channels. The tortoise has been demonstrated to be a good animal model for nociception. In this study, NAV 26 and A803467 selective blockers for Nav1.7 and Nav1.8 ion channels were used to investigate their ability to reduce nociception in the Speke's hinge back tortoise. The time spent in pain-related behavior during formalin, capsaicin and hot plate tests was recorded and analyzed. The results show that both blockers significantly reduced the mean time spent in pain-related behaviour in the formalin test and capsaicin test. Only A803467 caused a statistically significant decrease in the mean time spent in pain related behaviour in the hot plate test. In conclusion, Nav1.7 and Nav1.8 are significantly involved in processing of chemical induced inflammatory pain in the Speke's hinge back tortoise. However, only Nav 1.8 appears to participate in processing of thermally induced pain.

T12) TALK TITLE: **Acute Probiotic Treatment During the Perinatal Period Affects Maternal Care and Gut Microbiota Profiles of Long-Evans Rats**

AUTHORS: M. Elizabeth O'Leary, Elizabeth M. Myles, Isaac D. Romkey, Kiyana Kamali & Tara S. Perrot

ABSTRACT: In rodents, active maternal care is crucial for offspring; and a mother's behaviour is sensitive to her external environment. Therefore, the external environment can influence how a mother interacts with her offspring. It is essential to understand what impacts maternal care since low maternal care disrupts offspring development. Probiotics have the potential to alter the gut microbiota, and their downstream effects may impact maternal health and behaviour. We investigated the effects of probiotics on maternal care, offspring developmental outcomes and maternal gut microbiota. We found that probiotic-treated mothers performed increased amounts of active nursing and licking/grooming compared to their placebo-treated counterparts. At weaning, offspring from probiotic mothers weighed significantly more than those from placebo-treated mothers. Lastly, we found that probiotic-treated mothers' microbiota contained increased amounts of Bifidobacterium. These results contribute to our understanding of the external factors that may influence the mother, her health and behaviours and affect offspring outcomes.

T13) TALK TITLE: **Natural Photoperiodic Regulation of Hippocampal Plasticity in Wild Black-Capped Chickadees (*Poecile Atricapillus*)**

AUTHORS: Broderick M. B. Parks & Leslie S. Phillmore

ABSTRACT: In black-capped chickadees (*Poecile atricapillus*), food-storing behaviour fluctuates seasonally; supporting this behavioural change is a seasonal change in the hippocampus (Hp), critical for spatial behaviour and memory. Seasonal change in Hp is well-documented; we cannot however easily compare findings due to the use of captivity in some studies, and captivity is known to alter Hp plasticity. Also underlying this brain-behaviour interaction are the breeding-associated changes in avian physiology driven by photoperiod (day length). Here, we captured wild black-capped chickadees (N=48) across three distinct photoperiodic conditions: photostimulated (March-April), photorefractory (August-September), and photosensitive (December-January). All birds were sacrificed shortly after capture; we stained neural tissue with cresyl violet and quantified measures of Hp plasticity including volume and neuron number. Our findings are among the first to quantify seasonal Hp plasticity in a large sample of wild chickadees not introduced into any sort of captivity, and provides key evidence in assessing whether food-storing behaviour and its impact on Hp occur independently of photoperiod-driven seasonal regulation of breeding in food-storing birds.

T14) TALK TITLE: **Improvements to Rodent Housing Have Significant Effects on Development of the Stress Response: Implications for Modeling Neuropsychiatric Disorders**

AUTHORS: Tara Perrot

ABSTRACT: The value of rodents for modeling human neuropsychiatric disorders and unraveling underlying molecular mechanisms is not questioned. Attempts (not always successful) are made to standardize the conditions under which rats or mice are housed to facilitate comparing results generated in different labs all over the world. While the merits of standardizing housing seem obvious,

the costs are generally not considered. Over the last 10 years, we repeatedly have demonstrated significant effects of maintaining rats in semi-naturalistic housing, compared to standard housing, on the quality of maternal care. Decades of prior research has demonstrated the importance of maternal care quality for developing offspring, especially with respect to stress-related outcomes. Our work has revealed significant differences in stress-related brain and behaviour measures in offspring raised in semi-naturalistic housing versus standard housing. Here, I argue that standard housing is impoverished, and the conclusions drawn from results generated using standard housing should reflect that.

T15) TALK TITLE: **Is There One "Beam" of Attention for Searching in Space and Time?**

AUTHORS: Raymond M. Klein, Brett B. T. Feltmate, Yoko Ishigami & Nicholas E. Murray

ABSTRACT: In their pure forms, searching in space entails the allocation of attention to items distributed in space and presented at the same time whereas searching in time entails the allocation of attention to items distributed in time and presented at the same location. In independent projects we explored whether the metaphorical "beams" operating the domains of space and time might be independent or the same. In one project we used a differential approach. Early research using spatial (Snyder, 1972) and temporal search tasks (McLean, Broadbent & Broadbent, 1983) reported a substantial degree of sloppiness (binding errors). Our participants performed both of these tasks to see if the frequency of these binding errors in space and time were correlated. We replicated both early findings of binding errors in space and in time, but their frequency of occurrence in the two domains was not significantly correlated. In the other project, we used an experimental approach. With the participation of DISP students we explored whether the principles described by Duncan & Humphreys for searching in space would apply similarly to searching in time. Not surprisingly, performance in spatial search conformed to the predictions of D&H's principles. Importantly, temporal search performance followed the same pattern, suggesting that D&H's principles generalize to temporal search. Why do you think these two approaches yield different answers to the question posed in our title?

T16) TALK TITLE: **Pavlovian Conditioned Odour Memory in the *Neurexin1+/-* Mouse Model of Autism Spectrum Disorder**

AUTHORS: Jessica F. Garden, Ian C. G. Weaver & Richard. E. Brown

ABSTRACT: Neurexins (*NRXN*s) are presynaptic cell-adhesion molecules that bind with postsynaptic ligands to connect synapses and regulate synaptic transmission. Altered *Nrxn* gene expression is linked to neurodevelopmental disorders, such as autism spectrum disorder (ASD), in which learning and memory are impaired. The *Nrxn1+/-* transgenic mouse model, with the loss of one *Nrxn1* allele, is a novel model of ASD that is utilized in this research. Mice conditioned to dig in an odor pot to receive a reward during a 1-day Pavlovian conditioning paradigm using the method of Schellinck et al. (2001 Chem. Senses 26:663–72), were tested for short- and long-term olfactory memory. Given the recently identified role of *NRXN* in synaptic stability, we hypothesised that neurexin would be implicated in molecular processes required for long term memory, and thus, the *Nrxn1 +/-* mice would have impaired long-term odour memory. Preliminary analyses reveal that *Nrxn1 +/-* mice have reduced short-term memory during olfactory digging tasks compared to age- and sex-matched controls, and we expect to observe even greater deficits in long-term olfactory memory in *Nrxn1+/-* mice.

T17) TALK TITLE: Reconciling Expectation and Interpretation in the Cognitive Training Meta-Analyses

AUTHORS: Richard S. Drake, Gail A. Eskes & Raymond M. Klein

ABSTRACT: The cognitive enhancement potential of computerized, game-based working memory training tasks continues to attract researchers and investors alike. Laboratory and commercial endeavours of the past two decades have been both basic and applied, with goals ranging from understanding the mechanisms of learning and transfer to improving academic performance or reducing age-related cognitive decline. Hundreds of scholars have participated in the debate of whether working memory training is effective. Putting aside valid criticisms of the debated question itself ("effective" is not specific and is only meaningful when paired with clearly defined goals), there is evidence to support both sides. However, on the basis of an in-progress systematic review of the working memory training meta-analyses, we argue that the controversy has more to do with differences of interpretation, rather than numeric differences. We thus qualify the alleged quantitative controversy (or "quantroversy") accordingly. We will contextualize our preliminary findings with respect to transfer theories and recent best practice recommendations for cognitive training.

T18) TALK TITLE: Lexical Decisions in Ortho-Semantic Learning: A Behavioural Study of Grade 3 Children in English and French Immersion

AUTHORS: Laura Elliott, Alena Galilee, Lisa Beck, Clara Lownie, Catherine Mimeau, S. Helene Deacon & Aaron J Newman

ABSTRACT: English speaking children in grade 3, in English and French immersion, completed a novel ortho- semantic learning task (Mimeau et al., 2018). They read passages introducing novel words and their meaning, followed by a test on spellings and meanings of the words. Next, participants completed a lexical decision task (LDT) with the novel words, real words, non-words, consonant strings, and false fonts. While reaction times on the LDT for English students were significantly faster, accuracy did not differ significantly between groups. In the absence of a significant difference, French students demonstrated a trend towards higher accuracy for novel words, and lower accuracy for non-words. French immersion students receive formal reading education in a second language, and these results suggest it may reduce their speed for making judgements on English words. Furthermore, French students achieving reduced accuracy for rejecting non- words may reflect a less-established sight word vocabulary in English, compared to their English peers. However, heightened accuracy of French students for novel words provides a potential benefit of learning to read in a second language, as they become proficient in encountering unfamiliar words.

T19) TALK TITLE: The Visual Word Form Area as a Route of Auditory Linguistic Information Into the Visual Cortex of Early Blind Individuals: A Dynamic Causal Modelling FMRI Study

AUTHORS: K. O'Neil & A.J. Newman

ABSTRACT: Auditory language processing in the visual cortex of blind individuals is thought to be caused by input from the left frontal-temporal language network, however the pathways and brain areas involved are not established. The current fMRI study compared brain activity during semantic and phonological processing in early blind and sighted individuals. We used dynamic causal modelling (DCM) to determine how linguistic information reaches the visual cortex in blind individuals via effective

connections from the typical language network. Consistent with past studies, both early blind and sighted individuals showed activation in the frontal-temporal language network during semantic and phonological processing, including the inferior frontal gyrus (IFG) and the visual word form area (VWFA), but only early blind individuals demonstrated activity in the visual cortex (V2). DCM showed that in blind individuals, there was an endogenous connection from the VWFA to V2, and a direct modulatory connection from the VWFA to V2 during semantic and phonological processing. This suggests that in early blind individuals, the VWFA forms a reentrant endogenous connection to the visual cortex, and that visual cortex activity associated with semantic and phonological processing is the result of direct integration into the fronto-temporal language network, via connections with the VWFA.

T20) Concluding Remarks Aaron Newman

All Posters Will Be Presented From 12:30 pm to 2 pm
in the Lilyan E. White Study Centre on the 4th floor

POSTER TITLE: Perinatal Grief as a Mediator Between Religiosity and Sexual Satisfaction Among Couples Who Have Experienced Pregnancy Loss

AUTHORS: K. Brenna Bagnell, David B. Allsop & Natalie O. Rosen

ABSTRACT: Perinatal grief has been associated with declines in couples' sexual satisfaction. However, previous research has yet to address the potential factors associated with perinatal grief and sexual satisfaction such as religiosity. Researchers have found associations between religiosity, perinatal grief, and sexual satisfaction; however, no study has examined these associations together in a single model. Accordingly, the present study aimed to examine the extent to which perinatal grief mediates the association between religiosity and sexual satisfaction. A sample of 51 couples who had experienced a pregnancy loss within the last four months completed an online survey assessing their religiosity, perinatal grief, and sexual satisfaction. Analyses were conducted using a path analysis framework through the Actor Partner Interdependence Model of Mediation. I hypothesized that higher religiosity would be associated with lower perinatal grief, which would, in turn, be associated with higher sexual satisfaction. Contrary to my hypothesis, perinatal grief did not mediate the association between religiosity and sexual satisfaction. However, higher levels of the gestational partner's perinatal grief were associated with lower levels of their own and their partner's sexual satisfaction. These findings could inform interventions by suggesting that lowering perinatal grief might enhance sexual satisfaction after a pregnancy loss, ultimately leading to higher overall wellbeing.

POSTER TITLE: Investigating Retinal Inactivation as a Safe Therapy for Amblyopia

AUTHORS: Mitch Baker, Mairin Hogan, Nadia DiCostanzo, Tabitha O'Donnell, Andrew MacLean & Kevin Duffy

ABSTRACT: Disruption to normal vision during early life can produce a debilitating visual impairment called amblyopia, which is the most common cause of childhood blindness. The efficacy of conventional treatments for amblyopia are limited by a variety of factors including poor compliance with therapy,

recurrence after treatment, and poor recovery beyond early life. We developed a novel and more efficacious treatment for amblyopia that involves temporary retinal inactivation of the dominant eye with intravitreal microinjection of a local anesthetic. In this study we examined several characteristics of the visual system to investigate the safety of this novel treatment. Following inactivation, we observed no deviation from normal on the three measurements we made: (1) density thalamic neurons, (2) refractive error, (3) and body weight. Demonstrating the safety of inactivation therapy is a critical step in the translation of this application for human amblyopia.

POSTER TITLE: **The Power of the Pencil: Examining the Impact of Drawing on Learning and Memory**

AUTHORS: Briand, C., Klein, R., M. & Christie, J.

ABSTRACT: Drawing has recently been recognized as an aid to teaching, especially in the sciences. Through drawing, the student engages both their active cognitive system and motor system, and they are more self-reflective, thereby recognizing errors in their understanding of the material. While prior research suggests learning and memory are aided by these processes, few studies have examined the impact of drawing a dynamic visual scene; or simultaneously assessed memory and learning. The present study aimed to address these gaps in the literature, asking, compared to simply looking for an equal amount of time, does drawing a visual scene from observation differentially improve learning and memory?

POSTER TITLE: **Implications of Lactobacillus Probiotic Administration and Maternal Western Diet on Active Maternal Care and Offspring Development in Rats**

AUTHORS: Dante Cantini, Elizabeth O'Leary & Tara Perrot

ABSTRACT: Maternal obesity during pregnancy is associated with the development of adverse stress physiology in offspring. A maternal high-fat diet decreases the quality of active maternal care behaviours and increases offspring glucocorticoid levels in rats. Inadequate maternal care is associated with impaired development of the offspring's stress response. High-fat diets also negatively impact the intestinal microbiome and the gut-brain axis. Lactobacillus probiotics have been shown to ameliorate the intestinal microbiome in high-fat diet rats. Our study administered a Western diet to pregnant female rats. It investigated the impact of the Lactobacillus probiotic, Lacidofil®, on maternal care behaviour and offspring stress physiology. We hypothesized that probiotic administration to Western diet dams would mitigate the negative effects of a gestational high-fat diet. Behavioural scoring, ELISA, and RT-qPCR techniques were used to determine the effects of treatment. Few statistically significant effects were found; however, the results indicate that certain interactions and trending effects are occurring.

POSTER TITLE: **A Dyadic Comparison of Sexual Desire and Sexual Satisfaction Between Those Who Have Experienced Pregnancy Loss and Those Who Have Not**

AUTHORS: Eva Cohen, Natalie O. Rosen & David B. Allsop

ABSTRACT: Pregnancy losses are common and can negatively affect individual mental health and well-being. However, there has been little empirical research conducted on the effects of pregnancy loss on couples' sexual well-being. Accordingly, I examined the effects of a recent pregnancy loss on couples' sexual desire and sexual satisfaction. Couples that that had experienced a recent pregnancy loss and a community sample of couples that had never experienced a pregnancy loss completed online questionnaires about sexual well-being. Multilevel modeling was used to examine mean differences in self-reported sexual desire and sexual satisfaction among three groups. It was found that gestational partners (GPs) reported significantly lower levels of sexual desire than non-gestational partners (NGPs) and individuals of the community. It was also found that GPs reported lower average levels of sexual satisfaction than those of the community sample and that NGPs reported significantly lower levels of sexual satisfaction than individuals of the community sample. Results can provide evidence to aid in the development of interventions that aim to improve couples' levels of sexual desire and sexual satisfaction after a pregnancy loss. *Keywords:* pregnancy loss, sexuality, couples, well-being

POSTER TITLE: **Investigation of the Programming Effects of Probiotic Treatment on Metabolic and Anxiety-Related Outcomes in Male and Female Long-Evans Rats**

AUTHORS: L. M. Dauphinee, E. M. Myles & T. S. Perrot

ABSTRACT: Previous research has suggested positive health benefits on anxiety and obesity-related complications from administering specific strains of probiotics. Additionally, indirect probiotic exposure from pregnant mothers may metabolically program favourable health outcomes in young offspring. The purpose of this study was to characterize anxiety- and obesity- related effects of CEREBIOME® administration in male and female Long-Evans rats. The effects of CEREBIOME® were studied in two separate subsets of rats. Experiment A rats received CEREBIOME® or placebo from mothers throughout pregnancy and lactation, and directly from weaning until sacrifice. Experiment B rats were sacrificed at weaning and only received indirect CEREBIOME® or placebo exposure from mothers. In Experiment A, probiotic treatment significantly reduced certain anxiety-like behaviours in the Open Field Test. In Experiment B, the placebo group had significantly higher plasma leptin levels compared to the probiotic group. These results have implications for prevention and treatment of anxiety disorders and obesity-related complications.

POSTER TITLE: **The Role of Strategic Correction in Prism Adaptation Effects**

AUTHORS: Bayla Dolman, Caroline Rajda, Jasmine Aziz & Gail Eskes

ABSTRACT: Prism adaptation is a visuomotor learning paradigm and therapy for spatial neglect that involves wearing laterally right-shifting prisms to create reaching errors that stimulate learning and adaptation. There are mixed findings regarding whether strategic correction (SC; explicit error correction during prism exposure) inhibits, enhances, or is dissociated from post-prism adaptation resulting in leftward aftereffects. We aimed to investigate the temporal aspects of how SC influences adaptation and aftereffects. We manipulated young adults' SC by introducing a secondary task (letter-number sequence task) during early (trials 1-30) or late (trials 31-60) prism exposure to reduce SC and compared effects to a single task condition. Inhibiting SC during early prism exposure slowed error reduction, while inhibiting SC during late prism exposure did not impact error reduction. Aftereffects were unaffected by SC in either condition. While SC is important for initial error reduction, it does not appear critical for aftereffect development.

POSTER TITLE: **Personality, Stress, and Academic Cheating During the COVID-19 Pandemic**

AUTHORS: George R. Fazaa, Julie Blais & Luke R. Mungall

ABSTRACT: The quick spread of the coronavirus (COVID-19) pandemic has impacted academic institutions and students by forcing them to switch to online teaching and learning methods unexpectedly. Previous research has shown that students are more likely to cheat when stressed, and personality has also been shown to be a correlate of academic cheating. Thus, the current study examined the relationship between personality correlates (the HEXACO and the Dark Triad), stress due to COVID-19, and engagement in academic cheating during the Fall 2020 term (N = 535), while controlling for other established correlates (sex, age, GPA, and attitudes towards cheating). Our results indicated that being less honest and modest, more grandiose and manipulative, and more planful and deliberate were associated with more engagement in cheating. There was no effect clear effect of stress on cheating. We discuss the practical implications of the findings and future directions in academic cheating research.

POSTER TITLE: **The Tale of Two Antibodies: A NESC 3370 and NESC 3000 Project**

AUTHORS: Jonathon Henneberry, Nicholas Murphy, Meera Baskaran, Aoibh Crouse, Irene Euodia, Gabrielle Herzenberg, Alaa Alsmadi, Grace Clark, Rachel Gibbs, Omar Sabouni, Sophie Stewart, Jessica Telizyn & Kevin Duffy

ABSTRACT: Development of the visual system is characterized by an asymmetric capacity for neural plasticity in which neural circuits are readily modified by visual experience only early in life, during the critical period. The waning capacity for plasticity with age is thought to emerge as a consequence of an accumulation of proteins that inhibit plasticity, so-called molecular brakes. In this study we examined the developmental profile of a putative molecular brake in the visual cortex, semaphorin 3A, in an attempt to understand its potential role as an inhibitor of plasticity. Using two different antibodies against semaphorin, we found diametrically opposed developmental profiles. Whereas one profile indicated a progressive increase in semaphorin 3A with age consistent with it acting as a molecular brake, the second antibody presented a development profile that was completely opposite. Our findings underscore the need to express caution when interpreting immunolabeling results, and in the poster we suggest possible controls.

POSTER TITLE: **Automation Reliability and Its Influence on Trust Dynamics**

AUTHORS: Christopher Holland, Ben Rittenberg, Grace Barnhart & Heather Neyedli

ABSTRACT: Automation is at the forefront of many industries as a tool and aid to human counterparts to improve both productivity and quality. With human-automation teams, there is a need to understand how automation reliability affects user trust in the automated system. Both over-trust and under-trust can be detrimental to system performance; however, the temporal dynamics of trust with changing reliability level are relatively unexplored. The purpose of this experiment is to better define how user trust changes over time in conditions of increasing or decreasing reliability of automation. This experiment makes use of a dominant-colour identification task, in which users must decide whether a grid is mostly composed of blue or orange pixels. Automation provides a recommendation to the users, with the reliability of the recommendation either increasing or decreasing over time. Results from this experiment are expected to help better characterize the dynamics of trust in relation to changing reliability of the automation. It is anticipated that differences in trust will arise between the increasing and decreasing condition, likely caused by the intent of the human to use the automation recommendation in the event that the human identifies that it is reliable or unreliable.

POSTER TITLE: **Investigating the Impact of Pre-recorded Lecture Length on Mind Wandering and Learning Outcomes**

AUTHORS: Tess Konwisarz & Colin Conrad

ABSTRACT: It is a commonly held best practice to deliver lectures in short segments because it makes it easier to keep attention and prevent students' minds from wandering. However, it is difficult to assess the precise impact of mind wandering on knowledge retention from various formats of lectures because it is difficult to measure without disrupting a user's experience. Previous experiments have identified electroencephalography (EEG)-indexed correlates of mind wandering in online lecture contexts, allowing the passive measurement of mind wandering. This study explores the impact of pre-recorded online lecture design on MW by observing the differences in EEG signals corresponding to thought-probe reports of different MW varieties generated during different lecture length formats as well as the learning implications. Preliminary results show similar EEG markers of MW found in past studies, while possibly identifying additional ERP markers that collectively appear to distinguish between MW varieties in a pre-recorded lecture context.

POSTER TITLE: **Interaction of Memory and Attention in the Context of Directed Forgetting**

AUTHORS: Ryan Lewis & Tracy Taylor

ABSTRACT: This study aimed to determine whether magnified slowing of response times (RT) to a secondary probe following forget compared to remember instructions in an item method directed forgetting task are due to attentional withdrawal. A target search task consisting of pop-out and serial conditions was embedded in an item-method directed forgetting paradigm to control for task switching and manipulate levels of attentional demand. Results found search RTs were slower in serial search for different compared to same word-target location trials following remember but not forget instructions. Serial search revealed remember different trials to be slower than remember same trials and more

accurate than remember same trials and both forget conditions, being comparable to pop-out search accuracy. Remember instructions may lower attentiveness to location of a study item while encoding, decreasing accuracy of responses to a spatially overlapping target during attentively demanding searches despite faster response times.

POSTER TITLE: The Effect of Prosocial Behaviours on Preschooler's Accent Processing and Word Learning

AUTHORS: Madison B. MacLachlan, Yara Yazbek, Rebeka Workye & Drew Weatherhead

ABSTRACT: Studies suggest that morally salient information about a speaker may be linked with accent processing and word learning. Our study investigates whether presenting preschoolers with information about a group's prosocial behaviours impacts their comprehension of an unfamiliar accent and their likelihood to trust a member of the group in a selective word learning task. Participants are presented a story including either prosocial, neutral or antisocial information about a group of fictional characters who speak in an unfamiliar accent. Afterwards, participants are asked to correctly identify images corresponding to passages read by a member of the group. Next, participants must choose to endorse labels for novel objects from either a member of the group or an alternate character. Preliminary analyses show that preschoolers have slightly more difficulty understanding an unfamiliar accent than a familiar one and that they endorse the labels of the speaker below chance in the antisocial condition (.33).

POSTER TITLE: The Factor Structure of a New Executive Function Questionnaire in Young Adults: The Learning, Executive, and Attention Functioning (LEAF) Scale

AUTHORS: Munroe, C., Leckey, J., Johnson, S. & Jacques, S.

ABSTRACT: Executive functions (EFs) are cognitive processes implicated in goal-directed behaviour. Questionnaires originally developed to assess EFs in children have been adapted for adults (e.g., BRIEF-A; Roth et al., 2005), displaying good psychometric properties and predictive validity. However, these measures are expensive to administer, particularly to large samples. We investigated the factor structure of a new EF questionnaire – developed for children – in young adults. The Learning, Executive, and Attention Functioning (LEAF; Castellanos et al., 2018) Scale is free and comprises 11, 5-item subscales that measure components of EF, learning-related cognitive processes, and academic abilities. The LEAF was completed by 232 undergraduates (M_{age} = 20.63, SD = 2.45). Upon removing all non-EF items following a preliminary factor analysis, we performed exploratory factor analysis on all LEAF EF items. Results revealed a 7-factor solution that is highly concordant with the structure of the originally proposed LEAF EF subscales. LEAF EF items appear to assess seven, moderately correlated (.26 - .66) EF constructs, with 32 of 40 items factoring exactly as expected. The LEAF promises to be an alternative, accessible measure of EFs for young adults.

POSTER TITLE: **Is Gaze-Cuing More Like Endogenous or Exogenous Orienting?**

AUTHORS: Nicholas E. Murray, Richard S. Drake & Raymond M. Klein

ABSTRACT: Gaze-cuing is a form of visual orienting, in which a shift in one person's gaze triggers onlookers to orient in the same direction. It is unclear if uninformative gaze cues invoke endogenous (i.e., deliberate) or exogenous (i.e., reflexive) orienting. This study extended two past modifications of the Posner cuing paradigm which double-dissociated endogenous and exogenous orienting: in Experiment 1, participants were cued to search for a conjunction target amongst an array of similar distractor items. Past studies showed that exogenous orienting helps participants to make fewer illusory conjunctions. In Experiment 2, participants performed a two-alternative-forced-choice task where one target appeared more often than the other. Past studies showed that endogenous orienting decreases reaction time (a "non-spatial expectancy effect") for the likely target. We hypothesized that gaze-cuing would function exogenously (i.e., decrease illusory conjunctions but fail to enhance non-spatial expectancies) because gaze-cuing shares more properties with exogenous orienting than endogenous orienting.

POSTER TITLE: **How COVID-19 Restrictions Impacted Uptake of the Better Nights, Better Days Program**

AUTHORS: Myatt, A., Keys, E. & Corkum, P.

ABSTRACT: Pediatric sleep problems have been exacerbated by the COVID-19 pandemic. An implementation study of the *Better Nights, Better Days During COVID-19 (BNBD-COVID)* eHealth program was conducted to determine if this behavioural sleep intervention was helpful in reducing behavioural sleep problems during COVID-19. This sub-study (n = 442) examined how pandemic restrictions at baseline (e.g., school closures) as well as consequences of these restrictions (e.g., increased anxiety) influenced the number of BNBD program sessions completed (out of 5) in families who enrolled in the BNBD program. Both quantitative and qualitative data were collected from questionnaires and interviews with families who fully, partially, or did not complete the program. We found no significant relationship between the number of restrictions in place at baseline and overall session completion ($r = -.063$; $p = .203$). Significant relationships exist, however, between session completion and school closures ($r = -.112$; $p = .023$), increased anxiety ($r = -.116$; $p = .018$), decreased socialization ($r = -.097$, $p = .048$), and less of a bedtime routine ($r = -.102$; $p = .039$). Qualitative findings suggest that families with full engagement were less affected by COVID-19 restrictions, while families with less engagement were more affected. Many families were able to implement this eHealth program to address their child's sleep problems, although those most affected by COVID-19 may have had the most challenges implementing *BNBD-COVID*. This research will inform how the program is delivered to future families to improve implementation and sleep improvements.

POSTER TITLE: **Does Physical Activity Predict Executive Function in University Students?**

AUTHORS: Anika Nastasiuk, Jenn Leckey*, Shannon Johnson & Sophie Jacques *presenting author

ABSTRACT: Engagement in physical activity is associated with better executive function, specifically better inhibitory control and cognitive flexibility (De Waelle et al., 2021). The current study aimed to determine the association between different aspects of physical activity in the past month and executive

function in university students. Data were collected from 233 university students (179 female, 48 male, 5 non-binary, and 1 non-identified) with a mean age of 20.7 years. Physical activity was assessed by a self-report questionnaire based on the Canada Fitness Survey, and measured by the number of activities engaged in in the past month, as well as the total metabolic equivalent of task (MET) minutes in the past month. Executive function was measured by two self-report questionnaires; the Learning, Executive, and Attention Functioning (LEAF) scale (Castellanos et al., 2018), and the Barratt Impulsiveness Scale (BIS; Patton et al., 1995). Correlations were found between physical activity and overall executive function, as well as between physical activity and cognitive flexibility, but not inhibitory control. As well, correlations were found between physical activity and other aspects of executive function. Moreover, the number of different activities engaged in in the past month predicted executive function better than the total MET expended in physical activities. Results support previous research that found similar associations between physical activity and executive function, generally, and cognitive flexibility, specifically, but not inhibitory control. Future research should investigate the association between physical activity and executive function at multiple time points to determine the bidirectionality of this relation.

POSTER TITLE: **Examining "Immunity" From Attention Capture: Does Cueing Fixation Delay Disengaging From It?**

AUTHORS: Prasad, S., Feltmate, B. B. T.*, Briand, C., Sutherland, D. & Klein, R. M. *presenting author

ABSTRACT: In their seminal 1992 paper, Folk et al. demonstrated that when targets are identified by some feature (i.e., colour), participants could adopt an attentional control setting (ACS) such that only cues sharing that feature captured attention. Ruthruff & Gaspelin (2018; R&G hereafter), investigating whether participants could adopt an ACS for space, reported observing "immunity to attentional capture" by cues presented in "irrelevant" locations (e.g., locations wherein a target would never appear). Prasad, Mishra & Klein (2021) noted that in a design similar to R&G, Ishigami, Klein & Christie (2009) observed reduced capture, but not immunity. Prasad argued (and confirmed via a replication and extension of R&G) that this was due to Ishigami's inclusion of a warning beep on every trial, which equated alertness levels across cue conditions. R&G attempted to do the same in their E2 by including a centre cue condition, but Prasad argued that this introduced a new confound: cueing fixation might have delayed disengaging from it. The present study tested this hypothesis in two experiments closely modelled off of the designs of Ishigami and R&G.

POSTER TITLE: **Relationship Between Satisfaction and Willingness to Pay for the Better Nights, Better Days for Children with Neurodevelopmental Disorders Program**

AUTHORS: Silver, H., Orr, M., Corkum, P. & Weiss, S.

ABSTRACT: **Introduction:** The majority of children (up to 90%) with neurodevelopment disorders (NDD) experience sleep problems. The *Better Nights, Better Days for Children with Neurodevelopmental Disorders* program (*BNBD-NDD*) is a parent-guided eHealth behavioural sleep intervention for parents of children 4-12 years of age with ADHD, ASD, CP, or FASD. BNBD-NDD was developed to address access barriers to evidence-based care. Sustainability of eHealth interventions is an ongoing problem. A commercialization approach to sustainability is one potential solution in which parents would pay for the intervention from their own resources. **Objectives/Methods:** As part of the recently completed randomized controlled trial (RCT) evaluating the effectiveness of the *BNBD-NDD* program, data was collected about parents' willingness to pay for the intervention. In the current study, we examined parents' level of satisfaction with the *BNBD-NDD* intervention, and if higher satisfaction with the intervention predicted a higher willingness to purchase the program, and if satisfaction and willingness to purchase differed between diagnostic groups. We examined this data at baseline (before intervention) and 4-months post intervention. Participants included 45 parents who had completed the intervention and 4-month post-intervention surveys. **Results:** Overall, intervention participants show high satisfaction with the *BNBD*-NDD program, and on average, participants indicated on both baseline and 4-month questionnaires that they would be willing to pay above the envisioned $100 asking price for the intervention. Average ratings of satisfaction and willingness to purchase were stable across diagnostic groups. A small-to-moderate significant positive correlation was found between participant's level of satisfaction and their willingness to purchase at the 4-month follow-up ($r(45)=.312$, $p=.037$). Lastly, questionnaire responses indicated that offering in-person support alongside the eHealth intervention would increase the amount that participants would be willing to pay. **Conclusions:** The *BNBD-NDD* program is a step towards evidence-based eHealth interventions that can work to address barriers impacting access to care. The information gathered regarding parent satisfaction and willingness to purchase the program will help guide the future sustainability of the *BNBD-NDD* program.

POSTER TITLE: **The Impact of Anxiety Sensitivity on Self Esteem and Negative Affect Following a Statistics Test**

AUTHORS: Shephard, A., Workye, R., Cribbie, R. A., Flett, G. L., Alexander, S. M. & Mackinnon, S. P.

ABSTRACT: Anxiety sensitivity, a trait involving a tendency to misinterpret anxiety-related physical symptoms as indications of physical danger, serious illness, or humiliating social rejection, might amplify psychological distress following academic failures. We hypothesized that anxiety sensitivity will moderate the relationship between test difficulty and the outcomes, such that anxiety sensitivity results in greater decreases in self-esteem and increases in negative affect following a failed statistics test. This hypothesis was tested on 329 post-secondary statistics students, using a 2-group between-subjects design, where students completed either an "easy" or "hard" statistics test. Students in the hard-test condition experienced decreased self-esteem and increased negative affect. Further, anxiety sensitivity amplified this effect for social self-esteem, anxiety, dysphoria, and hostility; as anxiety sensitivity increased, the differences between test conditions also increased. As anxiety sensitivity is related to avoidant coping, awareness of this risk factor should help inform prevention strategies to promote student well-being and academic achievement.

POSTER TITLE: **The Role of Muscle in the Immune Response: A RT-Qpcr Study of Toll1 and Insulin Receptors in the Muscle and Fat Body of *Manduca Sexta***

AUTHORS: Zbarsky J., McMillan L.E., Zacher I. & Adamo S.A.

ABSTRACT: Animals respond to predator and pathogen attack by mounting systemic responses that span across multiple organs. Both anti-predator and immune responses require the consumption of scarce resources, leading to shifts in resources across different tissues in the caterpillar *Manduca sexta*. When both predators and pathogens are imminently threatening, the defense systems are reconfigured such that the net response is more than a simple summation of the immune and predator defense response. We examined the response of muscle and fat body in the caterpillar to an immune challenge,

mock predator exposure and a combination of both challenges. We measured changes in gene expression for Toll1 and insulin receptors using RT-qPCR. Results showed that Toll1 receptor gene expression is upregulated in fat body during both an immune challenge and during dual challenges. This result was expected as the fat body is important for the immune response and Toll1 receptors activate antimicrobial peptide production. However, Toll1 receptor expression was also upregulated in muscle during an immune challenge and dual challenges, suggesting that muscle also plays a role in the immune response. Insulin receptor expression also increased in both fat body and muscle during an immune challenge and during dual challenges, suggesting high metabolic activity in these tissues in response to these challenges. Muscle appears to play an active role in the response to a pathogen challenge through the increased expression of Toll1 and insulin receptors, supporting the concept of an integrated defense system.

POSTER TITLE: **The Effects of Cannabidiol Expectancy on Acute Measures of Stress and Anxiety**

AUTHORS: Zhekova, R., Perry, R., Endresz, N., Dockrill. K., Tanner. A., Spinella. T., Stewart, S. & Barrett, S.

ABSTRACT: In recent years, Cannabidiol (CBD) has been shown to attenuate anxiety- and stress-related responses in human participants. However, the extent to which these therapeutic effects result from non-pharmacological factors such as expectancy is unclear. The present study aimed to assess the independent effects of CBD expectancy on subjective and physiological measures of stress and anxiety using a sample of 31 volunteers (11 men). All participants consumed CBD-free hemp seed oil while assigned to one of two CBD-expectancy conditions (told CBD vs. told CBD-free). Following oil administration, participants completed a modified version of the Trier Social Stress Test. Marginal linear models revealed significant expectancy by time effects for heart rate following oil consumption, as well as significant time effects for heart rate and subjective measures of stress and anxiety. These results suggest that CBD expectancy alone may possess anxiolytic effects, thereby emphasizing the need to control CBD-related expectancies in future research.